

PROBING THE NEW IEEE RELIABILITY TEST SYSTEM (RTS-96): HL-II ASSESSMENT

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Abstract- This paper presents a set of investigations about the bulk reliability performance evaluation of the new *IEEE Reliability Test System (RTS-96)*. Several bulk reliability system indices representing a hierarchical level two (HL-II) assessment of the new system are provided. The full representation including three areas is analyzed and all results are based on AC flows evaluations considering corrective remedial actions. The assessment tool was the *NH-2* program.

Keywords: reliability, IEEE test-system.

I. INTRODUCTION

In 1979 the first version of the *IEEE Reliability Test System (RTS-79)* was proposed [1] by the former *Application of Probability Methods (APM) Subcommittee* of the IEEE Power System Engineering Committee. It was developed with the aim "to satisfy the need for a standardized data base to test and compare results from different power system reliability evaluation methodologies". Even at that time enhancements to the system were already expected to be incorporated to face special applications and new trends in the electricity production business. Accordingly, a second version of the *RTS* was proposed in 1986 expanding the data relating to the generating system [2]. After 1986, several other useful reliability test systems were also proposed and published [3-8] which helped to narrow the gap between the growing industry needs and available computational tools.

The former *APM-RTS Task Force* started the development of an enhanced test system (*RTS-96*) for use in bulk power

system reliability evaluation studies. It is expected that this new system will permit comparative and benchmark studies to be performed on new and already existing reliability evaluation techniques. The new *RTS-96* system was recently made public [9], and several members of the newly formed *Reliability, Risk & Probability Applications Subcommittee (RRPA)* suggested that it would be valuable to have a set of numerical bulk reliability assessments using existing and well known computational tools such as *COMREL*, *SICRET*, *TRELSS*, *GATOR*, and many others.

The objective of this paper is to contribute to this aim, presenting a *hierarchical level two* reliability assessment of the new *RTS-96*, obtained with the *NH-2 (3.5 release)* program. The interested reader will find a detailed description of this program in the literature [10,12]. It is emphasized that the paper presents only some of the most common and basic system indices found in the literature [15]. Load point indices are not discussed in the paper and the optional DC link is not taken into account. Only the three-area configuration is analyzed.

II. BRIEF ASPECTS OF RTS-79 versus RTS-96

The original *IEEE RTS* has 24 buses, 38 branches and a load of 2 850 MW. This system has proved to be quite useful and continues to be largely utilized by several researchers. However, some drawbacks have been pointed out such as the lack of data for DC links, interconnections, hydrological effects, system dynamic behavior, etc. Some of these deficiencies have been overcome in the proposed new *RTS*.

The new *IEEE RTS-96* actually provides single-area, two-area, and three-area configurations. The full system (including 3 areas) has 73 buses, 120 branches, and 96 generating units with a total generating capacity of 10 215 MW (3 405 MW in each area) for a system peak load of 8 550 MW (2 850 MW in each area). An optional DC link representation, and several kinds of data which were unavailable to the old *RTS* are also provided. Each one of the areas is identical to the original *RTS-79* (with the exception of sub-area 32). Figure 1 shows a simplified sketch of the

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new system for full configuration with three areas but not taking into account the optional DC link between areas 1 and 3.

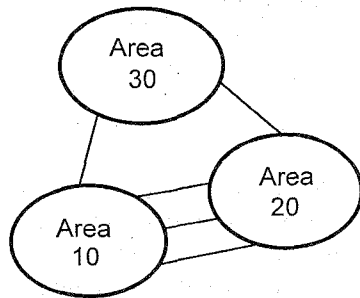


Figure 1- IEEE RTS-96 Sketch (3-area)

III. RTS-79 vs. RTS-96 OVERALL RELIABILITY

For the sake of comparison, Tables 1 and 2 present a set of reliability indices obtained for both the old and new systems. In this specific comparison only peak load condition was analysed. Transmission reliability indices (Table 1) were obtained by enumerating single and double circuit outages. Composite reliability indices were calculated by Monte Carlo simulation method. The relative uncertainty (β) shown in Table 2 was attained within 13 069 samples. The adequacy evaluation of each state was carried out through AC power flow model coupled to a successive linearization based remedial actions model. Violations caused by contingencies were eliminated by rescheduling of system generators, adjustments in voltage profile and taps of LTC transformers, and, if necessary, load curtailment. All these remedial actions are automatically activated in the program by a special code (referred as the *RDSQ* option: an acronym for the remedial actions mentioned previously).

The basic reliability indices are the loss of load probability (*LOLP*) and expectation (*LOLE*), expected energy not supplied (*EENS*), loss of load frequency (*LOLF*), loss of load duration (*LOLD*), and system problems probability (*SPP*). The later refers to the system condition before applying remedial actions. Some values of the *severity index* [11] are also occasionally presented.

Analysing the transmission reliability indices in Table 1, it is noted a decrease of the *EENS* in the new *RTS*, whereas the *LOLP* index has increased. As the new *RTS* is basically composed by three areas identical to the original *RTS*, we observe that the main circuit contingencies leading to load curtailment in the original system are still causing load

shedding in the new *RTS*. As their probabilities are summed up in the *LOLP* calculation, its value increases. On the other hand, in the new system, the severity of each load curtailment has decreased when the three systems are interconnected, leading to a decrease in the *EENS* index. Therefore, the transmission grid of the new *RTS* is slightly more robust than the original *RTS-79*, emphasizing the importance of interconnections. This shows that it is not pointless to compare the 3-area *RTS-96* with the 1-area *RTS-79*. Also, comparing the *SPP* and *LOLP* indices, it can be seen that the influence of remedial actions in the new system is quite important (efficient in 70% of cases).

In Table 2 it is again confirmed that the new test system is more reliable than the old one, since all global indices have shown improvements. For instance, the *EENS* of the old system is now 3.85 times higher than the value of the new *RTS*. These results confirm the benefits brought by interconnections for improving power systems reliability.

Still looking at Table 2, it is worthwhile to note that the old system shows a severity value (4 638 min) classified [11] as *degree 4*, which means "catastrophic consequences to consumers including system collapse or black-out". On the other hand, the new system has a severity reaching 401.34 min, classified as *degree 3*, meaning "very serious impact to all consumers". These results confirm that the traditional peak load bulk reliability analysis is quite pessimistic.

It is also interesting to assess the relative contribution of generation, transmission and composite events to the reliability indices. For example, for the old *RTS*, the generation, transmission and composite failures accounts to 77%, 3%, 20% and 76%, 1%, 23% for the *LOLP* and *EENS* indices respectively. For the new *RTS*, these contributions are 33%, 16%, 51% and 53%, 1%, 46%, for the same indices. Figures 2 and 3 illustrate these relative contributions on system *LOLP* for the old and new *RTS*, respectively. It is seen that the new *RTS* is still generation dominated, although its relative contribution has decreased. As a consequence, the relative contribution of composite failures has increased. Therefore, to better analysing the performance of the new *RTS*, a composite reliability tool is required.

Table 1 - Transmission Reliability Indices (Peak Load)

Indices	RTS-79	RTS-96
	Value	Value
SPP (%)	1.38	2.89
LOLP (%)	.35	.79
LOLE (h/year)	30.66	69.52
EENS (MWh/year)	1 053.14	622.02
LOLF (occ./year)	.74	8.34
LOLD (h)	41.16	8.33
Severity (min)	22.17	4.36

Table 2 - Composite Reliability Indices (Peak Load)

Indices	RTS-79		RTS-96	
	Value	$\beta(\%)$	Value	$\beta(\%)$
SPP (%)	25.31	-	7.1	-
LOLP (%)	15.12	3.57	4.92	3.84
LOLE (h/year)	1 324.92	3.57	431.66	3.84
EENS (MWh/year)	220 300.0	5.00	57 191.68	6.52
LOLF (occ./year)	27.31	9.23	25.47	6.74
LOLD (h)	48.50	8.46	16.94	5.50
Severity (min)	4 637.89	5.00	401.34	6.52

all results depicted in this section. In this figure, the load level probabilities and transition rates are also shown.

Two kinds of tests were envisaged to assess the RTS-96 performance. Both cases are compared with the set of results taken as reference (see *Case A*). The selected investigations are the following:

- Applying different types of remedial action (redispatching, load shedding, etc) to see effective in eliminating failure modes;
- Allowing wheeling transactions between critical buses;

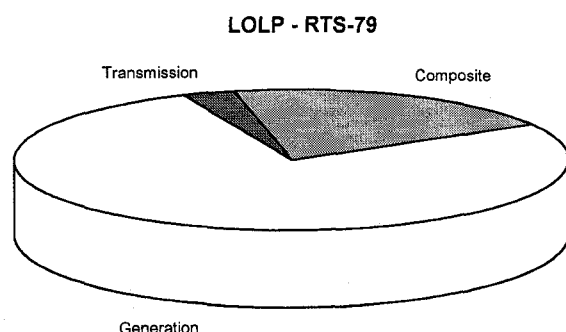


Figure 2- Relative Contribution of Failures on RTS-79

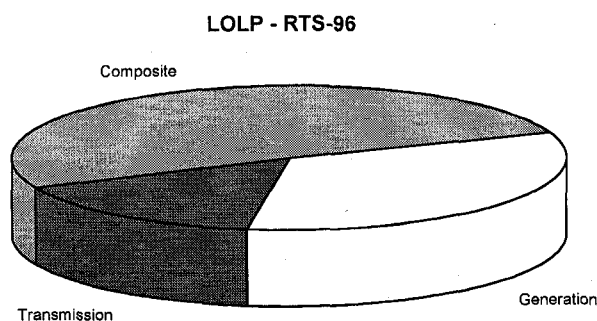


Figure 3- Relative Contribution of Failures on RTS-96

IV. RTS-96 RELIABILITY ASSESSMENT

In order to proceed with an accurate reliability assessment, the RTS-96 hourly load data (8760 values) was aggregated into 10 load levels, using the clustering techniques developed by Singh et al.[13,14]. It was found that choosing 10 equivalent load levels rendered a reasonable trade-off between accuracy and fast numerical evaluation. Figure 4 presents the load transition diagram obtained and utilized in

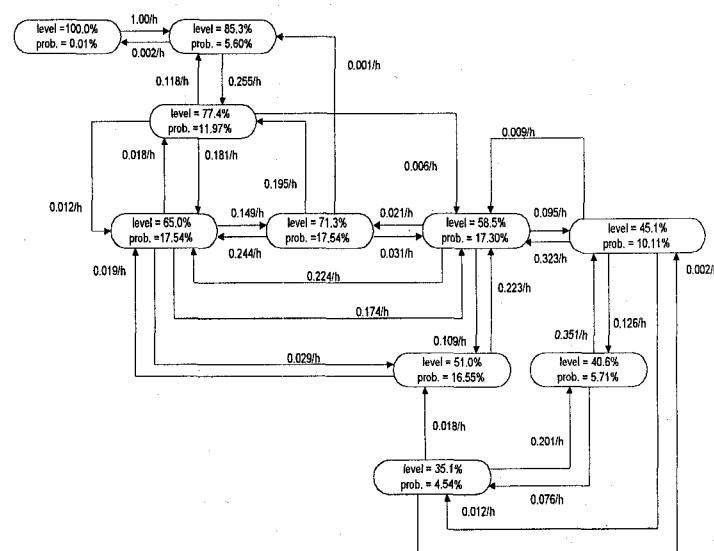


Figure 4- RTS-96 Load Transition Diagram.

CASE A: Benchmark Study

The results for this case (see Table 3) are taken as reference. Active and reactive redispatching as well as tap change in LTC transformers (*RDSQ*) are allowed to tentatively remedy the appearing failure modes.

The new system was submitted to both *state enumeration* and *Monte Carlo analysis*. The enumeration analysis does not take the generation system into account and considers *both single transmission and single + double transmission contingencies*. When single outages are analysed only 93.72% of the *transmission state space* is captured. When

single + double transmission contingencies are analysed, 99.24% of the *transmission state space* is covered.

The Monte Carlo assessment takes into account both generation and/or transmission contingencies allowing the evaluation of *composite reliability indices*. The simulations are performed for each one of the 10 load scenarios (see Fig. 4), obtaining conditioned reliability indices.

The final indices are calculated as the weighted sum of the conditional values, using the load level probabilities as weights. The total sample size encompasses 904 295 observations, optimally allocated among the load levels by using a stratification technique. A coefficient of variation β of 4.9% or even lower was attained, indicating that the evaluated reliability indices have a satisfactory precision level.

In Table 3, comparing the Monte Carlo evaluated SPP (2.03%) and LOLP (0.041%) indices one can see that the effectiveness of the remedial actions was 97.98% in this case. As suggested in [11], the obtained severity value (0.49 min) classifies the RTS-96 in *degree zero* (*system in acceptable normal condition*). The LOLP (0.041%) is equivalent to 1.5 days of failures in 10 years.

Table 3 - IEEE RTS-96 Benchmark Reliability Indices

Indices	Enumeration		Monte Carlo
	single	single+double	
SPP (%)	.17	1.70	2.03
LOLP (%)	.018	.021	.041
LOLE (h/year)	1.63	1.85	3.57
EENS (MWh/year)	3.46	6.28	69.98
LOLF (occ./year)	.46	.53	1.11
LOLD (h)	3.52	3.53	3.21
Severity (min)	.024	.044	.49

Regarding EENS, the comparison of the results of single + double enumeration (6.28 MWh/year) with Monte Carlo (69.98 MWh/year) indices emphasizes strikingly the importance of generation and composite failure modes.

Comparing Tables 2 and 3 one can also see the importance of careful load modeling. While the EENS related with the load peak study is 57 191.68 MWh/year, the same index drops 820.8 times to a value of 69.68 MWh/year when the load is represented with more detail (see Fig. 4).

The system areas reliability is shown in Table 4. It is interesting to observe that while the three areas are topologically almost identical (except for the nodes of interconnections), area A is significantly more reliable than the other two, both of which have approximately the same

reliability level. In all areas the 230 kV grid is stronger than the 138 kV.

Table 4 - Area Reliability

Area Identification	EENS (MWh/year)	Relative Contribution (%)
Area A (code 11), 138 kV	7.48	11
Area A (code 12), 230 kV	2.01	3
Area B (code 21), 138 kV	29.32	42
Area B (code 22), 230 kV	1.59	2
Area C (code 31), 138 kV	25.92	37
Area C (code 32), 230 kV	3.66	5
Total	69.98	100

Table 5 shows the failure modes relative contributions to the system reliability. It is seen that islanding and voltage violations are the two main failure causes.

Table 5 - Relative Contribution of Failure Modes

Failure Mode	Indices (Monte Carlo)				
	SPP	LOLP	LOLE	EENS	LOLF
Islanding	1.82	.013	1.13	47.8	.40
Overload	.025	-	.00031	.012	.00032
Voltage(over+under)	.16	.024	2.1	11.5	.58
Islanding + Voltage	.028	.003	.28	3.78	.089
Overload + Voltage	-	-	.0027	.0096	.0013
Others	-	.001	.058	6.89	.046
Total	2.03	.041	3.57	69.98	1.11

CASE B: Restricting Remedial Measures

Table 6 shows the results when the allowed corrective actions were restricted to only load shedding (SOCC, a program code that activates load shedding), as compared with the standard case (RDSQ, see section 3), where all controls can be optimized. All indices indicate system degradation, as expected. This results emphasizes the importance of remedial actions modeling in bulk reliability studies.

Table 6 - Influence of Different Remedial Actions

Indices Monte Carlo	Value (RDSQ)	Value (SOCC)
LOLP (%)	.041	.07
LOLE (h/y)	3.57	6.21
EENS (MWh/y)	69.98	215.6
LOLF (occ./y)	1.11	1.95
LOLD (h)	3.21	3.19
Severity (min)	.49	1.51

CASE C: Wheeling Transactions between Critical Buses

Table 7 shows the impact of a wheeling transaction on the system reliability levels simulated for peak load conditions (8550 MW).

As a by-product of the reliability analysis, expected values of sensitivity indices, measuring the variation on system energy not supplied with respect to variations on load increase were obtained. These sensitivities were used to selected sites for installation of new generation or loads. Bus 122 (Aubrey, 230 kV) was identified as the most sensitive RTS node to any system load increment, and therefore was chosen to be one of the buses involved in the wheeling transaction. Bus 204 (Bailey, 138 kV) was chosen to be the other one since it is in another electrical area, has a lower voltage level, has no generation, is radially connected, its sensitivity to load increase is negligible and therefore fits well to the purpose of showing the effects of a typical wheeling transaction.

In Table 7, three cases were analysed involving 10, 50 and 100 MW transactions from bus 122 to bus 204.

It is observed, that even modest transactions cause an impact to the overall system reliability indices. For instance, when the flow goes from bus 122 to bus 204, the EENS increases 6%, 9% and 13% for 10, 50 and 100 MW transaction, respectively. It is interesting to note that the other indices are still degraded, but not in the same proportion of EENS.

These results suggest that wheeling transactions, in general, should not be contracted without a previous comprehensive bulk reliability analysis of the entire power system [7].

Table 7 - Wheeling from Bus 122 to Bus 204

Indices Monte Carlo	Value case A	Value 10 MW	Value 50 MW	Value 100 MW
SPP (%)	7.1	6.9	6.76	6.92
LOLP (%)	4.92	4.87	4.86	5.03
LOLE (h/y)	431.66	426.49	426.06	440.80
EENS (MWh/y)	57 191.7	60 789.8	62 196.8	64 805.4
LOLF (occ./y)	25.47	26.86	27.48	29.02
LOLD (h)	16.94	15.88	15.50	15.19
Severity (min)	401.34	426.1	433.93	449.52

V. CONCLUSIONS

This paper has presented a preliminary reliability assessment of the newly released *IEEE RTS-96*. It was verified that the new system is more robust than the original *IEEE RTS-79*, in the sense that the reliability indices of the new system (see Table 3) are better.

The new *RTS-96* is classified as a *degree zero* system [11], and its overall LOLP is equivalent to 1.5 days of failures in 10 years.

It was also verified that in the new RTS the relative contribution of composite outages has increased when compared with the *RTS-79*. This result indicates that *the new RTS is better tuned for numerical bulk reliability assessments*.

Due to interconnections representation, the new system is also quite useful to reliability research involving wheeling transactions.

The authors suggest that, for the sake of comparison, the set of investigations presented in this paper be repeated by other researchers using other computational tools.

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